

Jonathan King

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SUMMARY

Natural Hazards Software Engineer and Computational Geoscientist. Designed and authored the USGS's operational postfire debris-flow hazard assessment ecosystem. Doctorate in climate data assimilation. Proven record of designing operational natural hazard software infrastructure, geospatial data pipelines, and numerical modeling tools.

SKILLS

Core Stack: Python, Bash/Shell, Git/Github, CI/CD (Github Actions), S3, uv, poetry, pytest, ruff, sphinx, typer, numpy

Geospatial: rasterio, pyogrio, pyproj, shapely, GDAL/OGR, xarray, geopandas, Spatial formats (GeoTIFF, COG, GeoJSON, Shapefile, GeoPackage, NetCDF, HDF5)

Engineering & Architecture: API Design, Package Architecture, Workflow Optimization, Data Schema Design, Open Source Distribution, CLI Tooling, High-Performance Numerical Computing

WORK EXPERIENCE

U.S. Geological Survey - Landslide Hazards Program

Lead Software Engineer / Mendenhall Fellow | Golden, CO | 2023 - present

- **Lead Software Architect:** Designed and built the operational postfire debris-flow (PFDF) hazard assessment ecosystem. Transitioned legacy workflows to modern, open-source production pipelines. Reduced compute times from hours to minutes.
- **Data Standards:** Wrote the data specification and schema for USGS PFDF hazard assessments, establishing standardized formats across agency pipelines
- **API Design:** Built *pwdf-api*, a Python and command-line API to search, query, and download USGS PFDF hazard assessments from public REST endpoints
- **Geospatial Libraries:** Designed and wrote the open-source PFDF software stack:
 - ***pwdf*:** A numerical Python library for debris-flow hazard analysis
 - ***wildcat*:** A command-line tool to produce and map PFDF hazard assessments
 - ***ocelote*:** USGS in-house codebase to produce and publish PFDF assessments

University of Arizona - Department of Geosciences

Computational Climatologist / Graduate Researcher | Tucson, AZ | 2016 - 2022

- **Software Engineering:** Built DASH, a MATLAB toolbox for paleoclimate data assimilation workflows

- **Geospatial Data Pipelines:** Engineered pipelines for coupling multi-model climate ensembles with observational products and paleoclimate proxy records
- **Scientific Impact:** Authored high-impact climate reconstructions in Nature, Nature Communications, Annual Reviews, and other scientific journals

EDUCATION

PhD, Climatology and Data Assimilation

University of Arizona | Tucson, AZ | 2022

MS, Climatology

University of Arizona | Tucson, AZ | 2017

BA, Chemistry and Applied Mathematics

Dartmouth College | Hanover, NH | 2015